

Skills Summary

Solving Right Triangles: Elevation and Depression

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding a side from an angle

Choose the ratio by what you have and what you want (SOH-CAH-TOA), with the angle measured from the horizontal:

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}.$$

If the unknown is on top, multiply. If the unknown is underneath, divide.

Example. A kite string 30 m long makes an angle of 55° with level ground. How high is the kite?

Step 1. Want the height, which is opposite the angle, and I know the string, which is the hypotenuse — opposite over hypotenuse is sine.

Step 2. height = $30 \sin 55^\circ = 24.5745 \dots$

height $\approx$ **24.57** m

Try it. An 8 m ladder makes an angle of 70° with level ground. How far up the wall does it reach?

check: 7.52 m

Skill 2 — Finding an angle from two sides

Rule. When two sides are known and the angle is not, use the inverse function of whichever ratio those two sides form:

$$A = \tan^{-1}\left(\frac{\text{opp}}{\text{adj}}\right), \quad A = \sin^{-1}\left(\frac{\text{opp}}{\text{hyp}}\right), \quad A = \cos^{-1}\left(\frac{\text{adj}}{\text{hyp}}\right).$$

Keep the calculator in degree mode.

Example. A ramp rises 3 m over a horizontal run of 20 m. Find its angle of elevation, to the nearest tenth of a degree.

Step 1. Want the angle and I know the rise (opposite) and the run (adjacent) — that pair is the tangent, so the angle is the inverse tangent.

Step 2. $A = \tan^{-1}(3/20) = 8.5307 \dots^\circ$

$A \approx$ **8.5** degrees

Try it. A ramp rises 4 m over a horizontal run of 25 m. Find its angle of elevation, to the nearest tenth of a degree.

check: 9.1 degrees

Skill 3 — Setting up a depression problem

Rule. An angle of *depression* is measured downwards from the horizontal at the observer. The observer's horizontal line and the ground are parallel, and the line of sight is a transversal, so

$$\text{angle of depression at the top} = \text{angle of elevation at the bottom.}$$

Redraw the depression angle at the *bottom* of the triangle and the problem becomes an ordinary right triangle.

Example. From the top of a 40 m cliff the angle of depression to a buoy is 22° . How far is the buoy from the foot of the cliff?

Step 1. Move the 22° down to the buoy, where it is the angle of elevation; then the cliff height is opposite it and the distance I want is adjacent — tangent, with the unknown underneath, so divide.

Step 2. $d = \frac{40}{\tan 22^\circ} = 99.0034 \dots$
 $d \approx \mathbf{99.00 \text{ m}}$

Try it. From the top of a 60 m cliff the angle of depression to a buoy is 35° . How far is the buoy from the foot of the cliff?

check 69.58 m

(!) Watch out: the angle of depression is never the angle at the top of the triangle between the vertical and the line of sight — that angle is its complement. Measure from the horizontal, every time.